Email: recruitment@iist.ac.in

Phone: 0471 2568422
Fax: 0471 256 8602

Government of India
Department of Space
Indian Institute of Space Science and Technology
(A Deemed to be University u/s 3 of the UGC Act, 1956)
Valiamala, Thiruvananthapuram - 695 547

सं. आईआईएसटी/प्रशा./No. IIST/Admn/RMT/4/14 /2015

प्रशासन / Administration
July 21, 2015.

Shri. Vijayakumar,
H.No. 11-1400-206-8,
Beside Reliance Tower, Biddapur Colony,
Behind Veertapaswi School,
Kalaburagi, Karnataka
Pin - 585 103.

Sir,

Sub: Admission to PhD programme July 2015

With reference to your application dated 29.04.2015 and subsequent interview you had with the Board of Selection on 26.06.2015, we are glad to inform you that you are selected for admission to the PhD programme July 2015 in IIST in the Department of Mathematics subject to the following terms and conditions.

(1) You will be paid a monthly fellowship of Rs.25,000/- (Rupees Twenty Five Thousand only) for the first two years subject to production of MSc. Degree Certificate.

(2) The admission will be subject to your being found medically fit by the competent medical authority. For this purpose you are advised to appear for a medical examination by the Medical Officer of Vikram Sarabhai Space Centre at Thiruvananthapuram on any working day (excluding Saturdays, Sundays and other declared holidays), after fixing up an appointment with Chief Medical Officer, VSSC (Tel No.0471-2562118/2563736). Alternatively, you may undergo the medical examination before the State Medical Authority closer to your place of stay. You are advised to ascertain your medical fitness before submitting the resignation letter to your present employer.

(3) You are required to take vaccination for Chickenpox and MMR before reporting for admission at IIST. In this respect a certificate is required to be produced in the attached format.

(4) The Doctoral programme in IIST begins with a course work, as stipulated by your Department on the basis of your research topic. The credit value for such course work will be decided based on your qualification.

(5) You will be required to assist the academic activities as well as sponsored research programmes of the Department, in addition to your regular research work, as per the needs of the Department.

...2/-

(6) After one year of joining, you will have to appear for a comprehensive examination which consisting of a written test, a Seminar and a Viva-voce. To continue with the Doctoral programme you will have to score a minimum of C grade in all the papers forming part of the course work. In case you fail to qualify the assessment, you may be given one final chance after a reasonable time, to be decided by the Doctoral Committee. No further opportunities will be available to qualify in the assessment and to continue with the Doctoral programme.

(7) After two years of research experience, your performance will be evaluated by a review committee constituted by the Director, IIST. Based on the review, the fellowship amount may be enhanced to Rs.28,000/-.

(8) The thesis will be evaluated by a specially constituted committee with experts drawn from outside, as per the guidelines issued by MHRD on the subject.

(9) You will have to pay tuition/registration fee and fee for the boarding/lodging/medical facilities etc., as per rules of the Institute from time to time.

(10) You are eligible for 30 day's leave of absence in a year, subject to a maximum of fifteen days in a semester including absence on medical grounds.

(11) You may be allowed to participate in relevant scientific events in India or abroad subject to approval by competent authority. Period of absence from the institute on account of participation in such approved events may be treated as on duty which is applicable to research scholars and the general rules of discipline applicable to all students.

(12) The disciplinary rules and students regulation of the Institute will be applicable to you during the course of your research in the institute.

(13) You are required to submit a detailed consolidated report of the research work done during the entire period of Fellowship on completion of the tenure/ resignation at the earliest.

(14) Your admission will be further subject to your producing the following documents at the time of admission:

(i) Documentary evidence regarding date of birth, nationality, educational qualifications, experience etc.
(ii) Transfer Certificate from the institution you have completed the post graduate programme.
(iii) An undertaking that you will not apply for any post or scholarship/fellowship elsewhere without the prior permission obtained in writing by the competent authority.
(iv) Four stamp-size recent colour photographs.

..3/-

(15) Please intimate us whether you are already under obligation to serve a central govt. dept./organization/state govt. /public sector undertaking. If so, you should produce a No Objection Certificate from the authorities concerned to your accepting this Offer of Admission.

If this offer for admission to Ph.D programme is acceptable to you on the aforesaid conditions, you are requested to communicate your acceptance to this office immediately and report to the Sr. Administrative Officer, Ph.D admissions. IIST on **July 27, 2015.**

Yours faithfully,

(A Chandrasekar)
Registrar

Copy to:

1. Deputy Registrar (Finance), IIST
2. Deputy Registrar (Academics), IIST
3. Personal file

cc: Director, IIST
Dean (R&D), IIST
Dean (Academics), IIST
Head of Department, Mathematics,, IIST

भारत सरकार / **Government of India**
अंतरिक्ष विभाग /**Department of Space**
भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान
Indian Institute of Space Science and Technology
वलियमला पोस्ट / **Valiamala Post**
तिरुवनंतपुरम / **Thiruvananthapuram - 695 547**

सं./ No. आईआईएसटी / IIST /प्रशा /Admn /SC15D021/15

प्रशासन /Administration,
अगस्त /August 20, 2015.

कार्यालय आदेश / OFFICE ORDER

श्री. विजयकुमार ने जुलाई **27, 2015** के पूर्वाह्न से अंतरिक्ष विभाग के भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान में (आईआईएसटी)₹25,000/-(रुपये पच्चीस हज़ार मात्र) प्रति माह की अध्येतावृत्ति पर गणित विभाग में पीएचडी शोध छात्र के रूप में कार्यारंभ किया हैं । उनको आबंटित छात्र कोड संख्या **SC15D021** है ।

Shri Vijayakumar has joined as PhD Scholar in the Department of Mathematics in Indian Institute of Space Science and Technology (IIST) of the Department of Space with effect from the forenoon of **July 27, 2015** on a monthly Fellowship of Rs. 25,000/-(Rupees Twenty Five Thousand only). He is allotted Student Code No. **SC15D021.**

अन्य सभी निबंधन एवं शर्तें, जुलाई 21, 2015 को उनको दिए गए आदेश सं. आईआईएसटी/प्रशा / 4/14/2015 में उल्लिखितानुसार होंगी और इसमें कोइ बदलाव नहीं होगा ।

All other terms and conditions will remain unchanged as specified in the offer No. IIST/Admn/RMT/4/14/2015 dated 21.07.2015 issued to him.

S. Ramanathan
20/08/2015

(एस. रामनाथन / S. Ramanathan)
वरिष्ठ प्रशासन अधिकारी / Senior Administrative Officer

सेवा में / To

श्री. विजयकुमार /Shri Vijayakumr - उचित माध्यम से /Thro Proper channel
छात्र कोड सं./Student Code No: **SC15D021**
पीएचडी शोध छात्र (गणित विभाग)/ PhD(Mathematics)
आईआईएसटी / IIST.

प्रतिलिपि /Copy to:

1) विभागाध्यक्ष, गणित विभाग / HoD, Mathematics, IIST
2) उप कुलसचिव (शैक्षिकी) / Dy. Registrar, Academics, IIST
3) उप कुलसचिव (वित्त) / Dy. Registrar (Finance), IIST
4) तंत्र प्रशासक व प्रबंधक / System Administrator cum Manager, IIST
5) वरिष्ठ प्रशासन अधिकारी, स्थापना / Sr. Administrative Officer, EST, IIST
6) वरिष्ठ प्रबंधक, होस्टल सेवाएँ / Sr. Manager–Hostel Services, IIST
7) वरिष्ठ प्रबंधक, कैन्टीन सेवाएँ / Sr. Manger-Canteen Services, IIST
8) पुस्तकालय अधिकारी / Library Officer, IIST
9) हिंदी अधिकारी/ Hindi Officer, IIST
10) मुख्य परामर्शदाता, सतर्कता एवं सुरक्षा / Chief Consultant (Vigilance & Security), IIST

P.T.O.

भारतीय अंतरिक्ष विज्ञान और प्रौघोगिकी संस्थान
तिरुवनतपुरम-695547

Indian Institute of Space Science and Technology
Thiruvananthapuram-695547, India

STUDENT IDENTITY CARD



VIJAYAKUMAR

Ph.D (Mathematics)

SC15D021

Valid upto **30-06-2020**

Issuing Authority

Address: H.NO-11-1400-206-8
BESIDE RELIANCE TOWER, BEHIND VEERTAPASWI SCHOOL
BIDDAPUR COLONY
KALABURAGI, KARNATAKA
Pin : 585103

Phone: 8105489930

Hostel : Chitra-205

Signature

D.O.B : 10-12-1989

Blood Group : A+ve

If found, please return to Director, IIST, Thiruvananthapuram - 695547.
P.R.O. : 91-471-2568453 Hostel : 91-471-2112021

101039

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY

Ph. D. PROGRAMME (Form 1)

REGISTRATION FORM

1. Name of the Ph.D. Scholar : VIJAYAKUMAR
2. Roll No. : SC15D021
3. Date of Birth : 10 Dec 1989
4. Category of Ph.D. Registration :

☑ Full Time		☐ Part Time
☐ IIST/ISRO Fellowship	☑ IIST Fellowship	DOS Centre/Unit.....

5. Department : MATHEMATICS
6. Date of Joining the Ph.D. Programme : 27 July 2015

---

7. Name of the Supervisor : Prof. Raju. K. George

   Name of the Co-supervisor and Dept. [if any]: Dept. of Mathematics
8. Brief statement of the proposed Ph.D. work :

   Mathematical theory of Control and Computational Schemes for steering controls.
9. Proposed course work for the entire Program:

Sl.No	Year	Semester	Course No.	Course Title	No. of credits
1	2015-16	Semn I	MA811	Research Methodology	03
2	2015-16	Semn I	AVC611	Mathematics for Control	03
3.	2015-16	Semn I	MA834	Advanced analysis	03
4	2015-16	Semn II	MA827	Functional analysis	03

10. Remarks of the HoD [Chairman, Doctoral Committee] :

11. Signature of Supervisor with date 12/10/15 Signature of Co-supervisor with date

12. Signature of HoD [Chairman, Doctoral Committee] with date:

Indian Institute of Space Science and Technology
(A Deemed to be University u/s 3 of the UGC Act 1956)
Government of India, Department of Space (DoS)
Valiamala P.O., Thiruvananthapuram 695 547, India
www.iist.ac.in

Prof. Raju K. George
Dean (R&D), IIST
dean-rnd@iist.ac.in
Tel: 0471 -2568413 Fax: 2568467

No: IIST: Dean (R&D): DC: **SC15D021:**2015

2nd November, 2015

OFFICE MEMORANDUM

Sub: Constitution of the Doctoral Committee for the PhD programme of Mr. Vijayakumar, SC15D021, IIST.

In accordance with PhD Rules and Regulations of the Indian Institute of Space Science and Technology and with the approval of the Director, IIST, the following Doctoral Committee is constituted for the PhD programme of **Mr. Vijayakumar,** SC15D021, Mathematics, IIST.

HoD, Mathematics, IIST	**Chairman**
Prof. M. P. Rajan, IISER, Trivandrum	**Member**
Dr. Anil Kumar Pundir, BITS Pilani, Goa Campus	**Member**
Dr. Priyadarshnam, IIST	**Member**
Dr. Deepak T. G., IIST	**Member**
Prof. Raju K. George, IIST	**Convener**

The Doctoral Committee will ratify the research outline and the specific course work identified for the scholar, after suggesting any modifications to the proposal from the Department, if required. After the PhD scholar successfully completes the prescribed course work, the Doctoral Committee will conduct the comprehensive examination of the scholar. The committee will continuously monitor and evaluate the progress of the research scholar and will advise on any midcourse corrections needed as per the Rules and Regulations of IIST.

Dean (R&D)

To
All Members of the Doctoral Committee

Cc
Registrar, IIST
Director, IIST

Indian Institute of Space Science and Technology
(A Deemed to be University u/s 3 of the UGC Act 1956)
Government of India, Department of Space (DoS)
Valiamala P.O., Thiruvananthapuram 695 547, India
www.iist.ac.in

Comprehensive Examination of PhD Scholars

Evaluation Report of the Doctoral Committee

Name of the PhD Scholar : Vijayakumar

Roll Number : SCI5D021

Department : Mathematics

Date : 16/08/16

Proposed Title of PhD work: Controllability and Observability of Nonlinear Systems

% Marks awarded by							Average %
Member1	Member2	Member3	Member4	Member5	Member6	Chairman	
75	73	78	76	75		80	76.17

% Marks obtained in the Written Exam(Comprehensive) : 75%

Overall Grade Awarded:

Remarks:

(Member 1) 16/8/2016 Prof M P RAJAN

(Member 2) Deepak. T. G

(Member 3) (ANIL KUMAR)

(Member 4) Priyadarshan

(Member 5) 16/8/16 (Raju K George)

(Member 6)

(Chairman)

Average Marks More than 80%:	OUTSTANDING
70% or above but less than 80%:	VERY GOOD
60% or above but less than 70%:	GOOD
40% or above but less than 60%:	AVERAGE
Less than 40%:	POOR

www.iist.ac.in

Indian Institute of Space Science and Technology
Govt. of India, Department of Space (DOS), Valiamala P.O,
Thiruvananthapuram - 695 547

DEPARTMENT OF MATHEMATICS

Minutes of the Ph.D Comprehensive Exam of Mr. Vijayakumar (SC15D021)

The Comprehensive Exam of Mr. Vijayakumar was held on 16th August 2016. The following members of the Doctoral Committee were present:

1.	Dr. K.S.S.Moosath, IIST	-	Chairman
2.	Dr. M.P. Rajan, IISER TVM	-	Member
3.	Dr. Anilkumar, BITS Goa	-	Member
4.	Dr. Deepak T.G, IIST	-	Member
5.	Dr. Priyadarshnam, IIST	-	Member
6.	Dr. Raju K George, IIST	-	Convener

After completing the Viva-Voce, Mr. Vijayakumar presented his research plan. Experts asked questions from the broad area of his research topic and an overall grade **"Very Good"** was given. The evaluation report is attached. By this he has completed all the requirements for registering his Ph.D program.

Dr. K.S.S. Moosath
(Chairman)
Doctoral Committee
IIST

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 11 Dated 16 Aug 2017

Approval for travel (airfare), guest house accommodation, transport and honorarium for DC outside expert

1. Name of the Ph.D Student : Vijayakumar
2. Student ID : SC15D021
3. Department : Mathematics
4. Category :IIST funded/~~External funded/Project funded~~/ ~~Internal IIST/Sponsored~~
5. Name of the Ph.D Supervisor : Prof. Raju K. George
6. Date of Enrolment for Ph.D : 27 July 2015
7. Broad research area : Control theory and it's applications

Members of Doctoral Committee (DC):

Sl. No.		Name	Designation and Email address
1.	Head of Department/DC Chairperson	Dr. K.S.S. Moosath	Professor smoosath@iist.ac.in
2.	Member-1	Dr. M.P. Rajan	Professor rajanmp@iisertvm.ac.in
3	Member-2	Dr. Anilkumar	Asst. Professor (on skype) anilpundir@goa.bits-pilani.ac.in
4	Member-3	Dr. Deepak T.G	Associate Professor deepak@iist.ac.in
5	Supervisor- Member Convener	Dr. Raju K George	Sr. Professor george@iist.ac.in
6	Member-4	Dr. Priyadarshnam	Associate Professor, priyadarshnam@iist.ac.in

The above-mentioned DC is meeting/~~met~~ on....17.08.17 at ....10..am

Director, IIST may kindly approve for travel (airfare), guest house accommodation, transport and honorarium for DC outside peer (subject) expert as per IIST norms.

Signature of Supervisor

Recommendation of HoD

Approved Director

Note: Approval from Director, IIST to be taken a week before the scheduled DC meeting

2801
22/5/17

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 7 Dated 17-08-2017

Approval of enhancement of PhD research fellowship

1. Name of the Ph.D Student : Mr. Vijayakumar
2. Student ID : SC15D021
3. Department : Mathematics
4. Category :IIST funded/~~External funded/Project funded/ Internal IIST/Sponsored~~
5. Name of the Ph.D Supervisor : Prof. Raju K. George
6. Date of Enrolment for Ph.D : 27 July 2015
7. Broad research area : Control theory and its applications

Members of Doctoral Committee

Sl. No.	Composition	Name	Designation and Email address
1.	Head of Department/DC Chairperson	Dr. K.S.S. Moosath	Professor smoosath@iist.ac.in
2.	Member-1	Dr. M.P. Rajan	Professor, rajanmp@iisertvm.ac.in
3	Member-2	Dr. Anil Kumar	Asst. Professor, anilpundir@goa.bits-pilani.a (On Skype)
4	Member-3	Dr. Priyadarshnam	Associate Professor, priyadarshnam@iist.ac.in
5	Supervisor- Member Convener	Dr Raju. K. George	Sr. Professor george@iist.ac.in
6			

The above-mentioned DC met on 17/08/17 at 10.am.. and have duly assessed the research progress of the above-mentioned PhD student based on interaction and the written progress report submitted by the PhD student.

Main Research Contribution by the student during the period of evaluation:
(indicated below or detailed Annexure enclosed)

To,
Deputy Registrar Academics

Overall Performance of candidate:

V. good

Final recommendation of DC:

DC is satisfied/~~not satisfied~~ with the research progress of the above research scholar and duly recommends/~~not recommends~~ the enhancement of fellowship after completing two years of PhD enrollment

Director, IIST may kindly approve the above recommendation of DC for enhancement of fellowship after completing two years of PhD enrollment

17/08/2017.
(Prof M.P Rajan)

17/08/2017
(PRIYADARSHNAM)

(K.S.S. Moosath)

Signature of DC members including DC Chairperson and Supervisor

Signature of HOD

Recommendation
(Dean R&D)

Approved
Director

22/8

Note: To be taken up after two years of completion after PhD enrollment

निदेशक का कार्यालय, भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान
Office of the Director, IIST
आवक सं./Inward No: 875
दिनांक / Date: 22/6/18

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 11

Dated 22 June 2018

Approval for travel (~~airfare~~), ~~guest house~~ ~~accommodation~~, transport and honorarium for DC outside expert

1. Name of the Ph.D Student : VIJAYAKUMAR
2. Student ID : SCISD021
3. Department : MATHEMATICS
4. Category : ~~IIST~~ funded/~~External funded~~/~~Project funded~~/ ~~Internal IIST~~/~~Sponsored~~

5. Name of the Ph.D Supervisor : PROF. RAJU K. GEORGE
6. Date of Enrolment for Ph.D : 27 JULY 2015
7. Broad research area : Controllability of impulsive & delay syste

Members of Doctoral Committee (DC):

Sl. No.		Name	Designation and Email address
1.	Head of Department/DC Chairperson	PROF. K.S.S. MOOSATH	smoosath@iist.ac.in
2.	Member-1	PROF. M. P. RAJAN	rajanmp@iisertvm.ac.in
3	Member-2	PROF. ANIL KUMAR	(~~through skype~~) anilpandis@goa.bits-pilani.ac.in
4	Member-3	PROF. DEEPAK. T. G.	deepak@iist.ac.in
5	Supervisor- Member Convener	PROF. RAJU K. GEORGE	george@iist.ac.in
6	Member- 4	PROF. PRIYADARSHAN	priyadarshanam@iist.ac.in

The above-mentioned DC is meeting/~~met~~ on 25 June 2018 at 9:30 am

Director, IIST may kindly approve for ~~travel (airfare), guest house accommodation~~, transport and honorarium for DC outside peer (subject) expert as per IIST norms.

22/06/18
Signature of Supervisor & Dean(R&D)

Recommendation of HoD

22.6.18

Approved
Director

22/6/18
डा वी के डढवाल
Dr V K DADHWAL
निदेशक, आईआईएसटी
Director, IIST

Note: Approval from Director, IIST to be taken a week before the scheduled DC meeting

डॉ. राजु के. जॉर्ज / Dr. Raju K. George
डीन (अनुसंधान व विकास) / Dean (R & D)
भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान
Indian Institute of Space Science and Technology
अंतरिक्ष विभाग, भारत सरकार
Dept. of Space, Govt of India
तिरुवनंतपुरम, Thiruvananthapuram - 695 547

निदेशक का कार्यालय, IIST तिरुवनंतपुरम
Office of the Director, IIST
आवक सं./Inward No. 1926
दिनांक / Date 26/6/18

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 6 Dated 25/06/18

Approval of recommendation of DC on the progress of PhD student

1. Name of the Ph.D Student : Mr. Vijayakumar M
2. Student ID : SC15D021
3. Department : Mathematics
4. Category : IIST funded/External funded/Project funded/ Internal IIST/Sponsored
5. Name of the Ph.D Supervisor : Prof. Raju K. George
6. Date of Enrolment for Ph.D : 22 July 2018
7. Broad research area : Control theory and its applications

Members of Doctoral Committee

Sl. No.	Composition	Name	Designation and Email address
1.	Head of Department/DC Chairperson	Dr. K.S.S. Moosath	Professor smoosath@iist.ac.in
2.	Member -1	Dr. M.P. Rajan	Professor rajanmp@iisertvm.ac.in
3	Member -2	Dr. Anil Kumar	Asst. Professor
4	Member -3	Dr. Priyadarshnam	Associate Professor Priyadarshnam@iist.ac.in
5	Supervisor- Member Convener	Dr. Raju K. George	Sr. Professor george@iist.ac.in
6	Member-4	Dr. Deepak T.G	Associate Professor deepak@iist.ac.in

The above-mentioned DC met on 25.06.18 at 9.45 am and have duly assessed the research progress of the above-mentioned PhD student based on the interaction and the written progress report submitted by the PhD student

Main Research Contribution by the student during the period of evaluation:
(indicated below or detailed Annexure enclosed)

Details Attached.

Overall Performance of candidate: Very Good

Final recommendation of DC: The committee evaluated the work and recommended to give pre-synopsis seminar by the end of July 2018 as he has fulfilled the norms of pre-synopsis seminar.

DC is satisfied/~~not satisfied~~ with the research progress of the above research scholar.

Director, IIST may kindly approve the above recommendation of DC

Deepak. T. G

(Raju K George)

(C PRIYADARSHANAN)
25/06/18

25/06/2018

Signature of all DC members including DC Chairperson and Supervisor

Signature of HOD 25.6.18

Recommended
26/06/18
Recommendation
(Dean R&D)

Approved
Director
26/6

Note: To be obtained annually after completion of Comprehensive examination

निदेशक, आईआईएसटी
Director, IIST

निदेशक का कार्यालय,
Office of the Director
आवक सं./Inward No.
दिनांक / Date: 28

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 11

Dated 28 June

~~Approval for travel (airfare), guest house accommodation~~, transport and honorarium for DC outside expert

1. Name of the Ph.D Student : VIJAYAKUMAR
2. Student ID : SC15D021
3. Department : Mathematics
4. Category : IIST funded/External funded/Project funded/ Internal IIST/Sponsored
5. Name of the Ph.D Supervisor : Prof. Raju K. George
6. Date of Enrolment for Ph.D : 27 July 2015
7. Broad research area : Control systems with impulses and d

Members of Doctoral Committee (DC):

Sl. No.		Name	Designation and Email address
1.	Head of Department/DC Chairperson	Dr. K.S.S. Moosath	Professor smoosath@iist.ac.in
2.	Member-1	Dr. M.P. Rajan	Professor rajanmp@iisertvm.ac.
3	Member-2	Dr. Anil Kumar Pundir	Asst. Professor (by skype) anilpundir@goa.bits-pilan
4	Member-3	Dr. Deepak T.G.	Professor deepak@iist.ac.in
5	Supervisor- Member Convener	Dr. Raju K. George	Sr. Professor george@iist.ac.in
6	Member-4	Dr. Priyadarshanam	Professor, priyadarshanam@iist.ac.in

The above-mentioned DC is meeting/met on 30.07.2018 at 10:30 am

Director, IIST may kindly approve ~~for travel (airfare), guest house accommodation~~, transport and honorarium for DC outside peer (subject) expert as per IIST norms.

28/06/18

Signature of Supervisor

डॉ. राजू के. जॉर्ज / Dr. Raju K. George
डीन (अनुसंधान एवं विकास) / Dean (R & D)
भारतीय अंतरिक्ष विज्ञान और प्रौद्योगिकी संस्थान
Indian Institute of Space Science and Technology
अंतरिक्ष विभाग, भारत सरकार
Dept. of Space, Govt. of India
तिरुवनंतपुरम / Thiruvananthapuram - 695 547

K.S.S. Moosath 25.6.18

Recommendation of HoD

Approved
Director

डॉ. वी. के. डढवाल
Dr V K DADHWAL
निदेशक, आईआईएसटी
Director, IIST

29/6/18

Note: Approval from Director, IIST to be taken a week before the scheduled DC meeting

भारत सरकार Government of India
अंतरिक्ष विभाग Department of Space
भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान
Indian Institute of Space Science and Technology
तिरुवनंतपुरम Thiruvananthapuram - 695 547

पीएचडी शोध प्रबंध प्रस्तुत करने हेतु प्रपत्र / Ph.D. THESIS SUBMISSION FORM

01. छात्र का नाम व छात्र पहचान सं. /Name of the Student & Student ID: Vijayakumar (SC15D021)
02. विभाग / Department : Mathematics
03. ई मेल आई डी व मोबाइल नंबर / Email ID & Mobile No. : vijayakumarmuni0@gmail.com
04. कार्यारंभ की तिथि / Date of Joining : 27 July 2015
05. सार संक्षेप प्रस्तुत करने की तिथि/ Date of Submission of Synopsis : 24 Aug 2018
06. शोध मार्गदर्शक (कों) का नाम / Name of Research Guide(s) : Prof. Raju K. George
07. पीएचडी शोध प्रबंध का शीर्षक, जैसे शोध प्रबंध में दिया गया है / Title of Ph.D. Thesis as given on the thesis:

Studies on Controllability of Dynamical Systems with Impulses and Time-Delay Controls

मैं एतद्वारा यह घोषणा करता / करती हूं कि मैं शोध प्रबंध प्रस्तुत करने की तारीख से एक महीने के अंदर होस्टल खाली कर दूंगा / दूंगी।
I hereby declare that I shall vacate the hostel within one month from the date of submission of thesis.

27 Dec 2018

छात्र के हस्ताक्षर व तिथि / Signature of the Student (with date)

[शोध मार्गदर्शक एवं विभागाध्यक्ष की सिफारिश]
[Recommendation by Research Guide & Head of the Department]

मैं / हम उपरोक्त शोध छात्र को पीएचडी शोध प्रबंध प्रस्तुत करने के लिए प्राधिकृत करता हूं / करते हैं।
I / We authorize the above scholar to submit the Ph.D. Thesis

Prof Raju K George

शोध मार्गदर्शक का नाम, तिथि व हस्ताक्षर
Name and Signature of Research Guide(s) (with date) 26/12/18

विभागाध्यक्ष के हस्ताक्षर व तिथि
Signature of HOD (with date) For HOD 27/12/18

डॉ. राजु के. जॉर्ज / Dr. Raju K. George
डीन (अनुसंधान व विकास) / Dean (Research & Development)
भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान
Indian Institute of Space Science and Technology
अंतरिक्ष विभाग / Department of Space, Govt. of India
तिरुवनंतपुरम, Thiruvananthapuram - 695 547

[बेबाकी प्रमाण पत्र / No Dues Certificate]

1. होस्टल सेवाएं / Hostel Services : No Dues in Hostel Services. 31/12/18.
2. पुस्तकालय सेवाएँ / Library Services : No Dues in IIST Library as on 27.12.2018 Asst. Librarian
3. विभाग / Department : No dues in Dept of Maths. Asst. Registrar (Academic) Indian Institute of Space Science and Technology Department of Space, Government of India Valiamala P.O. Thiruvananthapuram - 695 547 Hostel
4. लेखा - शोध प्रबंध प्रस्तुत करने का शुल्क रुपये **1000/-** एकत्रित करें
   Accounts - To collect Thesis submission Fee of **Rs.1000/-** 16178 31/12/18 R.K.V

जांच की और अग्रेषित/ Verified and forwarded
(उप कुलसचिव, शैक्षिकी / Deputy Registrar, Academics)

---

*मूल्यांकन के लिए पीएच.डी. शोध प्रबंध प्राप्त किया गया है और **स्वीकृत / अस्वीकृत** किया गया है।*
*The Ph.D. Thesis has been received and **Accepted/Not-Accepted** for evaluation*

डीन(शैक्षिकी) / Dean (Academics) 31.12.18

Government of India
Department of Space

Indian Institute of Space Science and Technology

Thiruvananthapuram 695547

Ph.D. Thesis Evaluation Report

1.	**Name of the Ph.D. Scholar**	: Mr. Vijayakumar S. Muni
2.	**Roll Number**	: IIST/DEAN AC/2018
3.	**Department**	: of Mathematics
4.	**Title of the Thesis**	: Studies on controllability of dynamical systems with impulses and time-delay controls

Recommendations

Tick one of the following:

1. The thesis in its present form is **commended** for the award of the Ph.D. Degree. ☑

2. The thesis is commended for the award of the Ph.D. Degree. However, my suggestions for improving the thesis are to be incorporated. At the time of the viva voce examination, the viva voce board has to ensure that the same are appropriately incorporated in the thesis. The revised thesis need not be sent to me. ☐

3. The thesis should be revised as per the suggestions enclosed. I would like to see the revised thesis incorporating my suggestions before I give further recommendations. ☐

4. The thesis is **not recommended** for the award of the Ph.D. degree. ☐

(Signature): J. Byszewski

Name of the Examiner: Ludwik Byszewski

Date: January 25, 2019

Please give your detailed report in the attached sheet highlighting your critical assessment of the work embodied in the thesis, any modifications/improvements to be made, the points to be clarified during the viva voce examination etc. You may use additional sheets, if required.

Ph.D. ~~Thesis Evaluation Report (contd.)~~ viva-voce

1. Name of the Ph.D. Scholar : Mr. Vijayakumar S. Muni
2. Title of the Thesis : Studies on controllability of dynamical systems with impulses and time-delay controls

~~DETAILED REPORT~~

Dear Professor George,

During the viva-voce, please ask Mr. V. S. Muni about:

(i) What is the networked impulsive system,

(ii) What is the class of bounded nonlinearities,

(iii) What is the class of Lipschitz nonlinearities.

Please give him excellent if he will give the answers for the above questions.

Kind Regards

Ludwik Byszewski

Jan. 25, 2019

(Signature): L. Byszewski

Name of the Examiner: Ludwik Byszewski

Date: January 25, 2019

Evaluation report of thesis

„Studies on controllability of dynamical systems with impulses and time-delay controls"

submitted to the Indian Institute of Space Science and Technology, Thiruvanthapuram-695 547, for the degree of Doctor of Philosophy in Mathematics

by

Mr. Vijayakumar S. Muni

The evaluated thesis consists in 7 chapters.

The thesis is about:

1. Introduction;
2. Preliminaries;
3. Controllability of semilinear impulsive control system with multiple constant time-delays in control;
4. Controllability of a fractional-order semilinear system with multiple constant time-delays in control;
5. Controllability of linear impulsive matrix Lyapunov ordinary differential systems with multiple constant time-delays in control;
6. Controllability of a linear impulsive system – an eigen value approach;
7. Controllability of a networked impulsive systems.

There are 118 important positions in Bibliography of the thesis and 4 positions in List of Publications by Mr. Vijayakumar S. Muni.

The evaluated thesis for Ph. D by Mr. Vijayakumar S. Muni is new, very interesting, original and excellent. The subject of the thesis is very difficult. The thesis is with congratulations.

The above Ph. D thesis is Highly Commended.

I support very strongly the application of Mr. Vijayakumar S. Muni to graduate his for the Degree of Doctor of Philosophy in Mathematics on the base of his Ph. D thesis.

Ludwik Byszewski

dr hab. Ludwik Byszewski, prof. PK
Institute of Mathematics
Cracow University of Technology
Warszawska 24
31-155 Cracow, Poland

January 25, 2019

Government of India
Department of Space

Indian Institute of Space Science and Technology

Thiruvananthapuram 695547

Ph.D. Thesis Evaluation Report

1. **Name of the Ph.D. Scholar :** Vijayakumar S. Muni
2. **Roll Number :** -
3. **Department :** Mathematics
4. **Title of the Thesis :** Studies on Controllability of Dynamical Systems with Impulses and time-delay Controls

Recommendations

Tick one of the following:

1. The thesis in its present form is **commended** for the award of the Ph.D. Degree. ☑
2. The thesis is commended for the award of the Ph.D. Degree. However, my suggestions for improving the thesis are to be incorporated. At the time of the viva voce examination, the viva voce board has to ensure that the same are appropriately incorporated in the thesis. The revised thesis need not be sent to me. ☐
3. The thesis should be revised as per the suggestions enclosed. I would like to see the revised thesis incorporating my suggestions before I give further recommendations. ☐
4. The thesis is **not recommended** for the award of the Ph.D. degree. ☐

(Signature):

Name of the Examiner: Prof. P. BALASUBRAMANIAM
Department of Mathematics
Gandhigram Rural Institute
(Deemed to be University)
Gandhigram (post), Dindigul-624302
Tamil Nadu, INDIA

Date: 4/3/2019

Please give your detailed report in the attached sheet highlighting your critical assessment of the work embodied in the thesis, any modifications/improvements to be made, the points to be clarified during the viva voce examination etc. You may use additional sheets, if required.

Ph.D. Thesis Evaluation Report (contd.)

1. Name of the Ph.D. Scholar : Vijayakumar S. Muni
2. Title of the Thesis : Studies on Controllability of Dynamical Systems with Impulses and Time-Delay Controls.

DETAILED REPORT

In this thesis, the necessary and sufficient conditions have been established for the controllability of dynamical systems in a finite-dimensional space on a continuous time-scale, for the following classes of impulsive and time-delay systems

i) Semilinear impulsive systems with multiple time-delays in control
ii) fractional-order systems of order $q \in (0,1)$ with multiple constant time-delay systems
iii) Systems modelled by Malvir Lyapunov ODEs possessing impulses and multiple delays in control
iv) Networked impulsive systems.

In chapter 3 & 4, Schauder's fixed point theorem has effectively used to derive the main results. In each chapter numerical example has been provided to validate the theoretical results. In general, the thesis is written well and more useful for further research.

(Signature):

Name of the Examiner: Prof. P. BALASUBRAMANIAM
Department of Mathematics
Gandhigram Rural Institute
(Deemed to be University)
Gandhigram (post), Dindigul-624302
Tamil Nadu, INDIA

Date: 4/3/2019

निदेशक का कार्यालय, आईआईएसटी
Office of the Director, IIST
आवक सं./Inward No. 1678
तारीख / Date 18/3/19

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 9 **Dated:** 15/03/19

Approval of the Ph.D Viva Voce Board and approval of travel, accommodation, transport and honorarium to Chairperson of the PhD Vive Voce Board

1. Name of the Ph.D Student : Mr. Vijayakumar Muni
2. Student ID : SC15D021
3. Department : Mathematics
4. Category : IIST funded
5. Name of the Ph.D Supervisor : Prof. Raju K George
6. Date of Enrolment for Ph.D : 27-07-2015
7. Date of Pre-synopsis seminar : 30-07-2018
8. Date of submission of Ph.D Thesis : 31-12-2018
9. Title of PhD thesis : Studies on Controllability of Dynamical Systems with Impulses and Time-Delay Controls

Recommendation of Ph.D Thesis Examiners :

Sl. No.	Name of Examiner	Email address of Examiner	Recommendation
1.	Prof. P.Balasubramaniam	balugru@gmail.com	Recommended W/O changes
2.	Prof. Ludwik Byszewski	lbyszews2pk.edu.pl	"

Director, IIST may kindly approve to constitute the Ph.D Viva Voce Board with the following composition.

Sl. No.	Designation	Name of Examiner	Email address of Examiner
1.	Chairperson -HOD/DC Chairperson	Dr. N. Sabu	n_sabu@iist.ac.in
2.	Member–Indian Examiner	Prof. P. Balasubramaniam	balugru@gmail.com
3	Member Convener-Supervisor	Prof. Raju K George	george@iist.ac.in
4			

The above-mentioned PhD Vive Voce Board is meeting/met on...29-03-2019 at .11:00am..

Director, IIST may kindly approve for travel (airfare), guest house accommodation, transport and honorarium for Indian Examiner of PhD Viva Voce Board as per IIST norms.

Signature of Supervisor

18/3/19
Signature of DC Chairperson

18/3/19
Signature of HOD

आईआईएसटी IIST DEPARTMENT OF MATHEMATICS

Recommendation
(Dean R&D)

18/3
Approved by Director

Dr. K. DADHWAL
निदेशक, आईआईएसटी
Director, IIST

Note: Composition of PhD Viva Voce Board to be completed within 10 days of receipt of reports from both the PhD examiners

निदेशक का कार्यालय, आइआईएसटी
Office of the Director,IIST
आवक सं./Inward No: 969
तारीख / Date: 02/4/19

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY
Valiamala P.O, Thiruvananthapuram

PhD Form 10 Dated: 29-03-2019

Approval of award of PhD degree

1. Name of the Ph.D Student : Mr. Vijayakumar Muni
2. Student ID : SC15D021
3. Department : Mathematics
4. Category : IIST funded
5. Name of the Ph.D Supervisor : Prof. Raju K George
6. Date of Enrolment for Ph.D : 27-07-2015
7. Date of Pre-synopsis seminar : 30-07-2018
8. Date of submission of Ph.D Thesis: 31-12-2018
9. Title of PhD thesis : Studies on Controllability of Dynamical Systems with Impulses and Time-Delay Controls.

Recommendation of Ph.D Thesis Examiners :

Sl. No.	Name of Examiner	Email address of Examiner	Recommendation
1.	Prof. Balasubramaniam	balguru@gmail.com	Recommended W/O Changes
2.	Prof. Ludwik Byszewski	lbyszews2pk.edu.pl	”

Ph.D Viva Voce Board members.

Sl. No.	Designation	Name of Examiner	Email address of Examiner
1.	Chairperson HOD/DC Chairperson	Dr. N. Sabu	n-sabu@iist.ac.in
2.	Member–Indian Examiner	Prof. P. Balasubramaniam	balguru@gmail.com
3	Member Convener - Supervisor	Prof. Raju K George	george@iist.ac.in

Overall Performance of the candidate while presenting Open-defense seminar:

Important Research Contributions by the student in his/her thesis: (indicated below or detailed Annexure enclosed)

Final Recommendation of the PhD Viva Voce Board:

The members of the PhD Viva Voce Board are satisfied/~~not satisfied~~* with the overall research contributions as well as responses provided by the research scholar regarding the clarifications raised by the PhD examiners and recommends/not recommends that Director, IIST approve the award of PhD degree to .....

(* the PhD Viva will need to repeated within the next three months)

29/3/19

Signature of Supervisor

Signature of DC Chairperson

Signature of HOD

29/3/2019

Signature of Indian Examiner - PhD Viva Voce board

2-4-19

Recommendation
Dean (Academics)

Vkdadhwal 2-4-19

Approved by Director

डॉ. वी. के. डडवाल
DR. V. K. DADHWAL
निदेशक, आईआईएसटी
Director, IIST

Note: Approval of Director, IIST to be taken within a week of the completion of the PhD open defense

भारतीय अंतरिक्ष विज्ञान एवं प्रौद्योगिकी संस्थान
(वि.अ.आयोग अधिनियम 1956 की धारा-3 के अधीन भावी मानित विश्वविद्यालय घोषित)
भारत सरकार, अंतरिक्ष विभाग, वलियमला पोस्ट, तिरुवनंतपुरम 695 547 भारत

Indian Institute of Space Science and Technology
(A Deemed to be University u/s 3 of the UGC Act, 1956)
Government of India, Department of Space
Valiamala P.O., Thiruvananthapuram 695 547 India

www.iist.ac.in

29th March 2019

DEPARTMENT OF MATHEMATICS

Ph.D Viva Voce Examination of Mr. Vijayakumar Muni

Ph.D Viva Voce exam of Mr. Vijayakumar Muni (Reg No: SC15D021) was held on 29th March 2019 at 11:00am in the Conference Hall (Aerospace block), IIST Trivandrum. Prof. P. Balasubramaniam, Gandhigram Rural Institute, the Indian Examiner of the Viva Voce Board was present in the viva-voce. Other members of the committee namely Prof. Raju K George, Dr. N. Sabu, were also present in the Viva Voce exam. Faculty members and students from IIST were present in the audience. Mr. Vijayakumar Muni presented his Ph.D research works and highlighted his contribution in the area of **"Studies on Controllability of Dynamical Systems with Impulses and Time-Delay Controls".** He has answered all the questions asked by the examiners and the audience. The viva-voce board is satisfied with the presentations and answers of the questions asked. The Viva Voce Board strongly recommends the award of Ph.D degree to Mr. Vijayakumar Muni by the Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram.

29/3/2019

Prof. P. Balasubramaniam
(Indian Examiner)

29/3/19

Prof. Raju K George
(Supervisor)

Dr. N. Sabu
(Chairperson)